

Операционные системы

Продвинутое использование git

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Получение навыков правильной работы с репозиториями git.

- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

```
aaharlamova@aaharlamova:~$ pnpm setup
Appended new lines to /home/aaharlamova/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/aaharlamova/.local/share/pnpm"
case ":$PATH:" in
  *:$PNPM_HOME:*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/aaharlamova/.bashrc
aaharlamova@aaharlamova:~$ source ~/.bashrc
aaharlamova@aaharlamova:~$
```

Рисунок 1: Node.js

```
aaharlamova@aaharlamova:~$ pnpm add -g commitizen

Update available! 10.27.0 → 11.25.0.
Changelog: https://pnpm.io/v/11.25.0
To update, run: pnpm self-update

[WARN] GET https://registry.npmjs.org/detect-indent error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
[WARN] Request took 29222ms: https://registry.npmjs.org/detect-indent
[WARN] GET https://registry.npmjs.org/merge error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
[WARN] Request took 29185ms: https://registry.npmjs.org/merge
Downloading @typescript/typescript-linux-x64@7.0.2: 9.67 MB/9.67 MB, done
[WARN] GET https://registry.npmjs.org/@typescript%2Ftypescript-netbsd-arm64 error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
[WARN] GET https://registry.npmjs.org/@typescript%2Ftypescript-darwin-arm64 error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
[WARN] Request took 29401ms: https://registry.npmjs.org/@typescript%2Ftypescript-netbsd-arm64
[WARN] Request took 29404ms: https://registry.npmjs.org/@typescript%2Ftypescript-darwin-arm64
[WARN] 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +148
+++++
Progress: resolved 167, reused 0, downloaded 148, added 148, done

/home/aaharlamova/.local/share/pnpm/global/5:
+ commitizen 4.3.2

Done in 1m 29.3s using pnpm v10.27.0
aaharlamova@aaharlamova:~$
```

Рисунок 2: установка commitizen


```
aaharlamova@aaharlamova:~$  
aaharlamova@aaharlamova:~$ pnpm add -g standard-changelog  
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6  
Packages: +14  
+++++  
Progress: resolved 181, reused 148, downloaded 14, added 14, done  
  
/home/aaharlamova/.local/share/pnpm/global/5:  
+ standard-changelog 8.1.0  
  
Done in 1.2s using pnpm v10.27.0  
aaharlamova@aaharlamova:~$
```

Рисунок 3: установка standard-changelog

```
1 {  
2   "name": "git-extended",  
3   "version": "1.0.0",  
4   "description": "Git repo for educational purposes",  
5   "main": "index.js",  
6   "repository": "git@github.com:arina-harlamova/git-extended.git",  
7   "author": "arina-harlamova",  
8   "license": "CC-BY-4.0",  
9   "config": {  
10     "commitizen": {  
11       "path": "cz-conventional-changelog"  
12     }  
13   }  
14 }
```

Рисунок 4: package.json

```
aaharlamova@aaharlamova:~/git-extended$ git add .
aaharlamova@aaharlamova:~/git-extended$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that don't modify src or test
  files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 3464b34] chore: test
1 file changed, 14 insertions(+)
create mode 100644 package.json
aaharlamova@aaharlamova:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 484 bytes | 484.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:arina-harlamova/git-extended.git
8993968..3464b34 master -> master
aaharlamova@aaharlamova:~/git-extended$
```

Рисунок 5: Отправка

```
aaharlamova@aaharlamova:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? []
Hooks and filters directory? [/home/aaharlamova/git-extended/.git/hooks]
aaharlamova@aaharlamova:~/git-extended$ git branch
* develop
  master
aaharlamova@aaharlamova:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/arina-harlamova/git-extended/pull/new/develop
remote:
To github.com:arina-harlamova/git-extended.git
 * [new branch]      develop -> develop
aaharlamova@aaharlamova:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

```
aaharlamova@aaharlamova:~/git-extended$ standard-changelog --first-release
aaharlamova@aaharlamova:~/git-extended$ git add CHANGELOG.md
aaharlamova@aaharlamova:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 4782a2e] chore(site): add changelog
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
aaharlamova@aaharlamova:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
Already on 'master'
Your branch is ahead of 'origin/master' by 2 commits.
(use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 1 +
1 file changed, 1 insertion(+)
create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was 4782a2e).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

aaharlamova@aaharlamova:~/git-extended$
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
aaaharlamova@aaaharlamova:~/git-extended$  
aaaharlamova@aaaharlamova:~/git-extended$ git push --all  
Enumerating objects: 6, done.  
Counting objects: 100% (6/6), done.  
Delta compression using up to 8 threads  
Compressing objects: 100% (4/4), done.  
Writing objects: 100% (5/5), 548 bytes | 548.00 KiB/s, done.  
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)  
remote: Resolving deltas: 100% (2/2), done.  
To github.com:arina-harlamova/git-extended.git  
 3464b34..e21f0c0  develop -> develop  
 3464b34..6eaf8fb  master -> master  
aaaharlamova@aaaharlamova:~/git-extended$ git push --tags  
Enumerating objects: 1, done.  
Counting objects: 100% (1/1), done.  
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.  
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)  
To github.com:arina-harlamova/git-extended.git  
 * [new tag]          1.0.0 -> 1.0.0  
aaaharlamova@aaaharlamova:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md  
https://github.com/arina-harlamova/git-extended/releases/tag/v1.0.0  
aaaharlamova@aaaharlamova:~/git-extended$
```

Рисунок 8: Отправка

```
aaharlamova@aaharlamova:~/git-extended$  
aaharlamova@aaharlamova:~/git-extended$ git flow feature start feature_branch  
Switched to a new branch 'feature/feature_branch'  
  
Summary of actions:  
- A new branch 'feature/feature_branch' was created, based on 'develop'  
- You are now on branch 'feature/feature_branch'  
  
Now, start committing on your feature. When done, use:  
  
    git flow feature finish feature_branch  
  
aaharlamova@aaharlamova:~/git-extended$ touch test2  
aaharlamova@aaharlamova:~/git-extended$ git flow feature finish feature_branch  
Switched to branch 'develop'  
Your branch is up to date with 'origin/develop'.  
Already up to date.  
Deleted branch feature/feature_branch (was e21f0c0).  
  
Summary of actions:  
- The feature branch 'feature/feature_branch' was merged into 'develop'  
- Feature branch 'feature/feature_branch' has been locally deleted  
- You are now on branch 'develop'  
  
aaharlamova@aaharlamova:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

aaharlamova@aaharlamova:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 757 bytes | 757.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:arina-harlamova/git-extended.git
   e21f0c0..e538265  develop -> develop
   6eaf8fb..5825844  master -> master
aaharlamova@aaharlamova:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:arina-harlamova/git-extended.git
 * [new tag]           1.2.3 -> 1.2.3
aaharlamova@aaharlamova:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/arina-harlamova/git-extended/releases/tag/v1.2.3
aaharlamova@aaharlamova:~/git-extended$
```

Рисунок 10: Завершение релиза

Подготовка рабочего репозитория

```
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ gedit package.json
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git add .
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git cz
cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:   Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 3ddf2ee] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 478 bytes | 478.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:arina-harlamova/2026-1--study--os-intro.git
   00852b6..3ddf2ee  master -> master
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 11: package.json и коммит

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged '1.0.0'
- Release tag '1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 728 bytes | 728.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To github.com:arina-harlamova/2026-1--study--os-intro.git
    3ddf2ee..824a396 develop -> develop
    3ddf2ee..4041186 master -> master
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:arina-harlamova/2026-1--study--os-intro.git
    * [new tag]          1.0.0 -> 1.0.0
aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.